

N=11 superpermutation of length 43,930,624

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October 4, 2026

1 Starting-phase optimization

Starting from the length-43,930,628 word found by Jakub Halfar, we obtained a verified 11-symbol superpermutation of length 43,930,624 (-4 savings).

We retained Jay Pantone's 2,800 closed trails, their cyclic transition sequences, and their order in the source word. We changed only their starting phases.

A starting phase specifies a position in the periodic spelling of a closed trail, including which constituent cycle is entered and the position within that cycle. From this position, we follow the fixed cyclic spelling until every permutation assigned to the trail has appeared as a complete window. This determines the shortest coverage-preserving opening for that position. The portion preceding the chosen start is encountered after wrapping around; it is not discarded.

For trail i , let c_i be its period length and P_i its set of admitted starting positions. For $p \in P_i$, define:

$$h_i(p) = \text{opened length} - c_i,$$

and let $F_i(p)$ and $T_i(p)$ be its first and last 11-symbol windows. Let $o(a, b)$ denote the maximal directed suffix-prefix overlap between a and b .

The assembled length is

$$L = \sum_{i=1}^{2800} c_i + \sum_{i=1}^{2800} h_i(p_i) - \sum_{i=1}^{2799} o(T_i(p_i), F_{i+1}(p_{i+1})).$$

We minimize this exactly by dynamic programming:

$$D_1(p) = h_1(p), \quad D_{i+1}(q) = h_{i+1}(q) + \min_{p \in P_i} [D_i(p) - o(T_i(p), F_{i+1}(q))].$$

Here $D_i(p)$ is the minimum accumulated opening adjustment minus overlaps for the first i trails, ending at opening p .

We admitted every permutation-window start assigned to its trail: 39,916,800 assigned occurrences plus 18,816 additional occurrences, giving 39,935,616 candidates. One opening was selected per trail. Exact suffix-prefix buckets evaluated the recurrence without constructing dense all-pairs overlap matrices.

Each opening preserves its assigned permutation set, and exact overlap joins preserve the opened words as contiguous substrings. The assigned sets partition S_{11} , so coverage is retained.

The result uses nine fewer opening characters but removes five fewer characters through overlap, saving four characters overall. Three independent full-word checkers confirmed all 11! permutations, with none missing.

This is a joint starting-phase optimization of the inherited construction, not a relocation of trails or a new standalone all- n construction.